

A fixed-weight disjoint-strict-singularity criterion for Orlicz–Lorentz inclusions

Candidate partial result for arXiv:2312.13903

12 August 2026

Status. Candidate partial result, likely valid. The theorem below settles a broad sufficient regime of the source problem and exactly characterizes a stronger uniform small-support property. It is not a full characterization of all DSS fixed-weight inclusions.

1 Source question

Bernal-González, Rodríguez-Vidanes, Seoane-Sepúlveda, and Tag ask in Section 5, item 4 of [1]:

What is the characterization of inclusion operators between Orlicz–Lorentz spaces to be disjointly strictly singular when the weight function w is fixed?

They emphasize that the Orlicz-space proof does not transfer plainly because of the weight and decreasing rearrangement. The exact source passage is shown below (source PDF page 34).

3. From the materials in Section 3.2.1, we retrieve the results in [9] and [33] if we let $\varphi(u) = u$.
4. From the results in Section 3.2, the following problem arises naturally: What is the characterization of inclusion operators between Orlicz–Lorentz spaces to be disjointly strictly singular when the weight function w is fixed? For Orlicz spaces, such characterization has been given in [22]. However, the proof for Orlicz spaces does not transfer to Orlicz–Lorentz spaces plainly because of technicalities arising from the weight function w and the decreasing rearrangement.
5. Using Theorem 3.7 and the problem posed at the preceding item, can we

2 Definitions and statement

Let w be a positive decreasing locally integrable weight on $[0, 1]$ and $W(t) = \int_0^t w(s) ds$. For an Orlicz function φ , write

$$\rho_{\varphi, w}(f) = \int_0^1 \varphi(f^*(s))w(s) ds, \quad \|f\|_{\varphi, w} = \inf\{a > 0 : \rho_{\varphi, w}(f/a) \leq 1\}.$$

The source says that φ_2 satisfies $\Delta_{\varphi_1}(\infty)$ when, for every $b > 0$, there is u_b such that

$$\varphi_1(u) \leq \varphi_2(bu) \quad (u \geq u_b). \tag{1}$$

Suppose the identity is a continuous inclusion $J : \Lambda_{\varphi_2, w} \rightarrow \Lambda_{\varphi_1, w}$. Define its small-support modulus by

$$\omega_J(t) = \sup \left\{ \frac{\|f\|_{\varphi_1, w}}{\|f\|_{\varphi_2, w}} : 0 \neq f \in \Lambda_{\varphi_2, w}, \ m(\text{supp } f) \leq t \right\}.$$

Theorem 1 (Exact uniform small-support criterion). *For Orlicz functions φ_1, φ_2 and a fixed weight w on $[0, 1]$, the following are equivalent:*

- (i) φ_2 satisfies $\Delta_{\varphi_1}(\infty)$;
- (ii) $\lim_{t \downarrow 0} \omega_J(t) = 0$.

Condition (i) itself guarantees that J is a continuous inclusion. Whenever these conditions hold, J is disjointly strictly singular.

Thus the growth condition completely characterizes a property stronger than DSS: the embedding constant vanishes uniformly on all sufficiently small supports. It is independent of the chosen fixed weight.

3 Proof intuition

Normalize a function in the domain norm and split its decreasing rearrangement at one amplitude A . Relative Δ controls the high amplitudes after asking for the condition at the scale $b = \eta/2$; convexity then supplies a factor $1/2$. The low amplitudes contribute at most a fixed constant times $W(t)$, which vanishes with the support size. This proves the uniform estimate. Conversely, normalized characteristic functions read off the ratio of the inverse Orlicz functions exactly, and that inverse ratio is equivalent to relative Δ . Disjoint supports in $[0, 1]$ necessarily shrink, so the uniform estimate rules out an isomorphic disjoint restriction.

The unresolved part is necessity for DSS itself. If uniform decay fails, characteristic blocks have individual norm ratios bounded below, but arbitrary linear combinations of those blocks undergo a new decreasing rearrangement. That prevents the individual estimates from producing an isomorphism on the whole disjoint span.

4 Proof

We first record the elementary inverse-function form of relative Δ :

$$\varphi_2 \in \Delta_{\varphi_1}(\infty) \iff \lim_{s \rightarrow \infty} \frac{\varphi_2^{-1}(s)}{\varphi_1^{-1}(s)} = 0. \quad (2)$$

Indeed, if (1) holds and $s = \varphi_1(u)$, then for all large u , $\varphi_2^{-1}(s) \leq bu = b\varphi_1^{-1}(s)$. Since $b > 0$ is arbitrary, the ratio tends to zero. Conversely, if the ratio tends to zero, then for a fixed $b > 0$ and all large u , $\varphi_2^{-1}(\varphi_1(u)) \leq bu$, which is (1).

Assume (i), and fix $0 < \eta < 1$. Apply (1) with $b = \eta/2$, obtaining a threshold u_0 . Put $A = \eta u_0/2$. If $r \geq A$, then convexity and $\varphi_1(0) = 0$ give

$$\varphi_1(r/\eta) \leq \frac{1}{2}\varphi_1(2r/\eta) \leq \frac{1}{2}\varphi_2(r). \quad (3)$$

Choose $\delta > 0$ so small that

$$\varphi_1(A/\eta)W(\delta) \leq \frac{1}{2}. \quad (4)$$

This is possible because $W(\delta) \downarrow 0$.

Take $0 \neq f \in \Lambda_{\varphi_2, w}$ with $m(\text{supp } f) \leq \delta$, and put $g = f/\|f\|_{\varphi_2, w}$. The Luxemburg normalization gives $\rho_{\varphi_2, w}(g) \leq 1$, while $g^*(s) = 0$ for $s \geq \delta$. Splitting where $g^* < A$ and $g^* \geq A$, and using (4) and (3), yields

$$\begin{aligned} \rho_{\varphi_1, w}(g/\eta) &= \int_0^\delta \varphi_1(g^*(s)/\eta)w(s) ds \\ &\leq \varphi_1(A/\eta)W(\delta) + \frac{1}{2} \int_0^\delta \varphi_2(g^*(s))w(s) ds \\ &\leq \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

Therefore $\|f\|_{\varphi_1, w} \leq \eta\|f\|_{\varphi_2, w}$. Since η was arbitrary, $\omega_J(t) \rightarrow 0$. The same relative condition implies $\varphi_1 \prec_\infty \varphi_2$, so continuity of J also follows from Theorem 3.7 of [1].

Conversely, assume (ii). The characteristic-function calculation (Lemma 3.1 of [1]) gives

$$\|\chi_{[0, t]}\|_{\varphi, w} = \frac{1}{\varphi^{-1}(1/W(t))}.$$

Hence

$$\frac{\|\chi_{[0, t]}\|_{\varphi_1, w}}{\|\chi_{[0, t]}\|_{\varphi_2, w}} = \frac{\varphi_2^{-1}(1/W(t))}{\varphi_1^{-1}(1/W(t))} \leq \omega_J(t) \rightarrow 0.$$

Because W is continuous and strictly increasing from zero, $1/W(t)$ ranges through all sufficiently large values as $t \downarrow 0$. Thus (2) holds, proving (i).

Finally, let (f_n) be any sequence of nonzero pairwise disjoint functions in the domain. The support measures tend to zero because their sum is at most one. Condition (ii) therefore gives

$$\frac{\|Jf_n\|_{\varphi_1, w}}{\|f_n\|_{\varphi_2, w}} \rightarrow 0.$$

If J were an isomorphism on the closed span of (f_n) , the displayed ratios would have a uniform positive lower bound. This contradiction proves that J is DSS. $\square$

5 The opposite endpoint and examples

Corollary 2. *If $\varphi_2 \in \Delta_{\varphi_1}(\infty)$, the fixed-weight inclusion is DSS. If instead both $\varphi_1 \prec_\infty \varphi_2$ and $\varphi_2 \prec_\infty \varphi_1$, the two Orlicz–Lorentz spaces have equivalent norms and the inclusion is not DSS.*

Proof. The first assertion is Theorem 1. In the second case, Theorem 3.7 of [1] applies in both directions, so the identity is an isomorphism on the whole infinite-dimensional space and hence on the span of any disjoint sequence of nonzero characteristic functions. $\square$

For example, with $\varphi_1(u) = u^p$ and $\varphi_2(u) = u^q$, $1 \leq p < q < \infty$, the inclusion $\Lambda_{q, w}[0, 1] \hookrightarrow \Lambda_{p, w}[0, 1]$ is DSS for every fixed weight w . If the two generating functions are equivalent at infinity up to dilations, the inclusion is not DSS.

6 Verification, limitations, and novelty status

The proof uses only four checkable facts: Luxemburg normalization, $f^* = 0$ beyond the measure of $\text{supp } f$, convexity $\varphi(\theta u) \leq \theta\varphi(u)$ for $0 \leq \theta \leq 1$, and the exact characteristic-function norm. No numerical evidence or ordinary Δ_2 hypothesis is used.

This is a partial answer. The implication “DSS $\Rightarrow \Delta_{\varphi_1}(\infty)$ ” is not claimed. Failure of the small-support condition produces disjoint functions whose individual norm ratios stay positive, but that does not control all scalar combinations after decreasing rearrangement. The full oscillatory/intermediate-growth regime therefore remains open.

A bounded literature search on 12 August 2026 used arXiv:2312.13903, the exact title, and combinations of “fixed weight”, “Orlicz–Lorentz”, and “disjointly strictly singular”. It located the source, the classical Orlicz/Lorentz results cited there, Astashkin’s general symmetric-space gap theory [2], and later variable-Lebesgue work, but no explicit later solution of the fixed-weight question. Astashkin’s results make the criterion conceptually adjacent to known gap theorems, and it may be implicit in broader almost-compact embedding theory; originality confidence is therefore moderate rather than high.

Human-review recommendation. Check the inverse-growth equivalence (2), the normalization in the high/low split, and whether a general almost-compact embedding theorem already states this fixed-weight criterion. Subject to those checks, the partial theorem should be retained as likely valid.

References

- [1] L. Bernal-González, D. L. Rodríguez-Vidanes, J. B. Seoane-Sepúlveda, and H.-J. Tag, *New Results in Analysis of Orlicz–Lorentz spaces*, arXiv:2312.13903; Adv. Math. 489 (2026), 110808.
- [2] S. V. Astashkin, *Disjointly strictly singular inclusions of symmetric spaces*, Math. Notes 65 (1999), 3–12, DOI:10.1007/BF02675003.